

HumanEgo: Zero-Shot Robot Learning from Minutes of Human Egocentric Videos

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Project page: <https://humanego-ai.github.io/>

Code: <https://github.com/TX-Leo/HumanEgo>

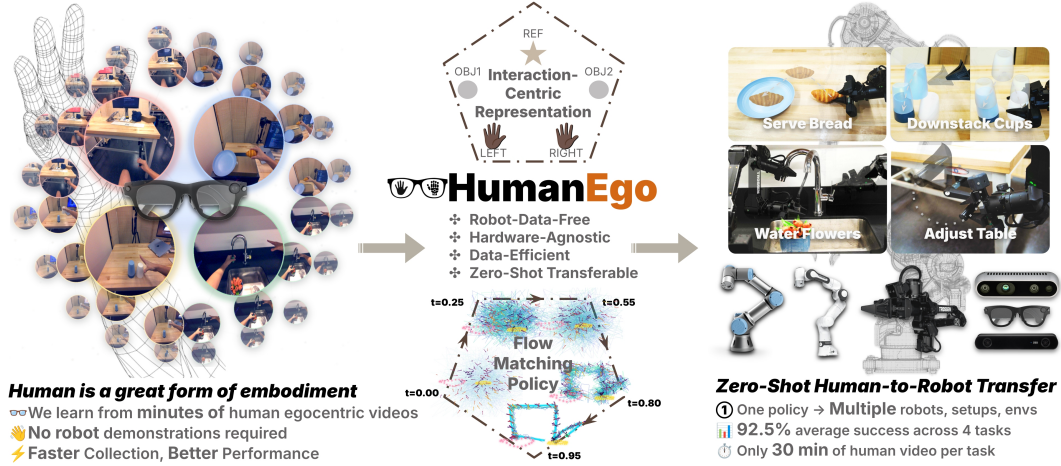


Fig. 1: **HumanEgo learns robot policy from human egocentric videos.** A human wears Aria glasses and collects demonstrations (left); the egocentric videos are converted into an interaction-centric representation and used to train a flow matching policy (middle); the policy transfers *zero-shot* to the robot—free of environment, setup, or embodiment (right).

Abstract: Human egocentric video captures rich manipulation demonstrations without any robot hardware, yet transferring these skills to robots remains challenging due to the embodiment gap between human and robot in both visual appearance and kinematics. We present **HumanEgo**, a framework that bridges the embodiment gap by lifting each human demonstration to an entity-level representation of *hand-object interaction*, and training a *flow matching* policy with *dense auxiliary objectives* that amplify supervision from every trajectory. HumanEgo is robot-data-free, hardware-agnostic, data-efficient, and zero-shot human-to-robot transferable. With only 30 minutes of human videos per task, HumanEgo achieves **92.5%** average success across four real-world tasks (**75%** with just 15 minutes), outperforms matched-time robot teleoperation by **41** percentage points, and robustly transfers zero-shot across novel robots, cameras, and environments. We release HumanEgo as an easy-to-use, open-source framework for learning robot policies directly from human data: <https://github.com/TX-Leo/HumanEgo>.

Keywords: Human-to-Robot Transfer, Egocentric Videos, Efficient Learning

1 Introduction

State-of-the-art manipulation policies require hundreds to thousands of task-specific robot demonstrations [1–6], which are costly, time-consuming, and inconvenient to collect. Human egocentric video offers a much cheaper and more accessible alternative: with a head-mounted camera [7], a single person can collect task demonstrations anywhere, in minutes. But how should we leverage this data? Existing approaches fall into two paradigms, each with significant limitations. *Co-training* methods [8–11] supplement robot data with human video, but still require substantial robot demonstrations for every new task—reducing, rather than eliminating, the data burden. *Large-scale pre-training* approaches [12–15] learn from massive egocentric corpora, but demand enormous compute and still require robot-specific post-training to produce deployable policies. We pursue a more direct goal: **learning deployable manipulation policies from only minutes of human egocentric demonstrations—without any robot data or internet-scale pretraining.**

Achieving this goal exposes two fundamental challenges. **(1) The representation challenge: bridging the embodiment gap.** Humans and robots differ in both *visual appearance* and *kinematics*, and these gaps demand distinct solutions. On the visual side, retargeting-based methods [16–18] synthesize robot-like imagery from human video but are brittle to morphological and viewpoint differences; point-tracking approaches [19–21] extract sparse geometric features but discard the rich visual context surrounding interactions. On the kinematic side, hierarchical methods [22–24] separate high-level plans from low-level execution but still require robot data for the low-level controller; object-centric approaches [25–27] track only the manipulated object, losing critical information about *how* the hand approaches, grasps, and releases it. We argue that neither hand nor object alone defines a skill—what matters is their *interaction*. This is the key representational claim behind HumanEgo: robots should not imitate the human body, but recover the task-relevant *interaction geometry* that transfers across bodies.

(2) The learning challenge: learning from minimal data. Although raw human video is abundant online, clean clips with precise action labels remain scarce, making data-efficient learning from minutes of per-task videos critical. This regime introduces two distinct challenges: *multi-modality* and *signal sparsity*. For the multi-modality challenge, the same task admits many valid strategies. Diffusion-based methods [3] capture this distribution but need many denoising steps and are slow at inference; faster alternatives [28] are less expressive. For the signal-sparsity challenge, each trajectory carries rich signal beyond the hand action—object motion, visual traces, hand–object state—yet prior work taps only a fraction: single auxiliary targets such as visual foresight [25, 29, 30] or 2D tracks [19, 27, 31], or upstream pretraining corpora [12, 13, 32]. We argue that a fast generative policy paired with multi-type dense supervision is the key to data-efficient learning from minutes of human egocentric videos. In other words, the goal is to squeeze many forms of supervision from human video, so small curated demonstrations can punch above their size.

We present HumanEgo (Fig. 1), addressing each gap with a targeted design. For the *visual gap*, we inpaint the human arm from each egocentric frame and render a virtual gripper with tracked object keypoints in its place, producing an embodiment-agnostic visual observation. For the *kinematic gap*, we encode every hand and object as an Interaction-Centric Token (ICT), producing a compact, embodiment-invariant spatial observation whose hand–entity blocks are viewpoint-invariant by construction. For *multimodality*, we adopt a flow matching [33] policy, producing expressive multi-modal actions at fast inference. For *signal sparsity*, we design three dense auxiliary objectives: 2D trace, object motion, and latent consistency. Together they produce multi-type dense supervision from each trajectory’s scene dynamics, boosting learning from few demonstrations. Our contributions:

- **HumanEgo**, a robot-data-free, hardware-agnostic, and data-efficient pipeline that learns robot manipulation policies from minutes of raw human egocentric videos—powered by a transferable interaction representation and a flow matching policy with dense auxiliary objectives.
- **Interaction-Centric Tokens (ICT)**, a compact entity-level representation of hand–object interaction that is embodiment-invariant, with hand–entity blocks that are exactly viewpoint-invariant.
- **Robust zero-shot human-to-robot transfer.** Trained on 30 minutes of human video per task, HumanEgo reaches **92.5%** success across 4 real-world tasks and **75%** at half that budget. At matched collection time, it surpasses robot teleoperation by **41** percentage points. The learned policy also deploys zero-shot to novel robot embodiments, camera setups, lighting, backgrounds, and object instances, without any retraining or fine-tuning. This suggests that human video is not merely a cheap substitute but a *scalable* and *potentially superior* data source for policy learning.

2 Related Work

Large-scale egocentric and hand–object interaction datasets [14, 15, 34–39] underpin learning manipulation from human video. Generalist policies and world models [12, 13, 40–42] learn embodiment-agnostic representations from massive corpora, but demand heavy compute and per-task robot post-training; co-training [8–11, 26, 43, 44] and visual retargeting [16–18, 23] still need robot demonstrations or renderings brittle to morphology and viewpoint; and hierarchical methods [22, 24, 45] delegate low-level control to a robot-trained controller. Zero-shot methods avoid

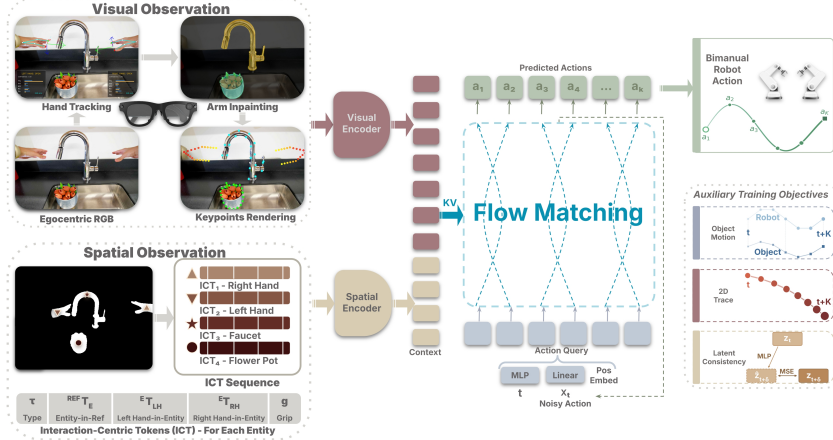


Fig. 2: **System overview of HumanEgo.** Arm inpainting and visual keypoints bridge the visual gap; Interaction-Centric Tokens encode spatial relationships among all entities; a flow matching policy with dense auxiliary objectives learns bimanual robot actions from minutes-scale human data.

robot data entirely and differ in *what* they represent: points [20, 21, 46, 47], object pose or motion [29, 31, 48–50], or image goals [19, 51], with further directions explored in [25, 30, 52–57]. All encode the hand *or* the object, rarely their *interaction*—the very signal that defines manipulation. HumanEgo encodes it explicitly, achieving zero-shot transfer from minutes of human video without robot data or large-scale pretraining. Appendix A expands this discussion.

3 HumanEgo

HumanEgo turns human egocentric video into a deployable bimanual policy in five stages (Fig. 2). A demonstrator wearing Aria glasses records the task (Sec. 3.1); we close the *embodiment gap* by inpainting the human arm and rendering a virtual gripper (Sec. 3.2) and by encoding every entity’s pose relative to other task entities into Interaction-Centric Tokens (Sec. 3.3). A flow matching policy with three auxiliary objectives generates multi-modal bimanual actions (Sec. 3.4), and the same policy drives a real robot by swapping only the modules that produce its inputs (Sec. 3.5).

3.1 Egocentric Data Collection

A human demonstrator wearing Aria Gen1 glasses [7] performs the target task in any convenient environment—regardless of table height, lighting, or background, and without specialized workspace or calibration. Demonstrations are recorded at 30 Hz. Aria glasses are particularly well suited for learning from human video: their Machine Perception Services (MPS) provide high-quality 6-DoF SLAM tracking, calibrated 3D hand pose estimation, and synchronized egocentric RGB streams—all from a single lightweight wearable device.

3.2 Visual Observation Preprocessing

We transform the undistorted egocentric frames into embodiment-agnostic RGB observations in two steps. First, we segment the human hand and arm with SAM2 [58] and remove them via LaMa inpainting [59], eliminating the visual embodiment gap. Second, we render a virtual gripper and the tracked object keypoints into the inpainted image—both derived from the spatial observation (Sec. 3.3)—implicitly encoding 6D pose information as visual cues. This lightweight procedure bridges the visual embodiment gap without expensive domain adaptation or image translation.

3.3 Spatial Observation Preprocessing

We build our explicit entity-level spatial observation: treating every object and both hands as an *entity*, we track the hands and objects to recover each entity’s 6-DoF pose, then encode their relative relations into Interaction-Centric Tokens. We detail these three steps below:

Hand tracking and motion optimization. We start from the 3D hand keypoints produced by Aria MPS [7], lift them to the world frame via SLAM, and smooth them with Savitzky–Golay on positions and an exponential moving average (EMA) on rotations. We then treat the thumb–index pair as a

virtual parallel-jaw gripper, extracting an $SE(3)$ end-effector pose T_{ee} and a scalar grasp g . Position is the fingertip midpoint; orientation is a Gram–Schmidt frame built on the metacarpophalangeal (MCP) joints rather than the fingertips, which avoids the degeneracy when fingertips converge during pinch grasps; and the grasp scalar $g \in [0, 1]$ normalizes the thumb–index fingertip distance, binarized at deployment (Appendix E.3).

Object tracking and pose estimation. We detect each object with text-prompted Grounding DINO [60], segment it with SAM2 [58], and sample contour keypoints from the mask. We track these 2D keypoints $\mathbf{u}_n$ across the video with CoTracker3 [61] and lift them to 3D via $\mathbf{p}_n = \text{Triangulate}(\mathbf{u}_n, K, T_{\text{SLAM}})$, using camera intrinsics K and the per-frame Aria SLAM pose T_{SLAM} . We take the centroid of the N tracked points as the object position to cancel per-point triangulation noise, $\mathbf{p}_{\text{obj}} = \frac{1}{N} \sum_{n=1}^N \mathbf{p}_n$, and estimate orientation R_{obj} with Orient-Anything V2 [62]. During grasping the object is occluded by the hand, so we apply *kinematic latching*—rigidly tying the object pose to the hand from the grasp onset t_0 : $T_{\text{obj}}^t = T_{\text{hand}}^t \cdot (T_{\text{hand}}^{t_0})^{-1} T_{\text{obj}}^{t_0}$. The latch replaces an estimate that occlusion has made unreliable with one propagated by the hand, and assumes the object does not move relative to the hand once grasped.

Entity Spatial Encoding via Interaction-Centric Tokens (ICT). We encode each entity’s 6-DoF pose into an ICT, capturing both its pose in a shared reference frame and its spatial relation to both hands. For each entity $k=1, \dots, N$, the token $\text{ICT}_k \in \mathbb{R}^{29}$ is:

$$\text{ICT}_k = \left[\underbrace{\tau}_1 \parallel \underbrace{\text{REF}T_E}_9 \parallel \underbrace{E_{T_{LH}}}_9 \parallel \underbrace{E_{T_{RH}}}_9 \parallel \underbrace{g}_1 \right], \quad (1)$$

where τ is the entity type (hand or object); $\text{REF}T_E$ is entity k ’s pose in a reference frame REF that is held fixed for the episode—the camera pose at the onset of manipulation during training, where the head-mounted camera moves, and the static camera itself at deployment (Sec. 3.5); $E_{T_{LH}}$ and $E_{T_{RH}}$ are the left-hand (LH) and right-hand (RH) poses expressed in entity k ’s local frame E ; and g is the grasp state (binarized finger distance for hands; a sentinel for objects). We flatten each $SE(3)$ transform to a 9D vector by concatenating the normalized translation with a 6D rotation representation [63], and derive every quantity from off-the-shelf perception without ground-truth labels. Unlike prior methods using global point clouds or absolute coordinates [20, 21], we anchor each ICT to an entity, so that $E_{T_{LH}}$ and $E_{T_{RH}}$ track the hand–object configuration as it evolves through approach, grasp, and transport, making the representation *interaction-centric* by construction. Expressing them relative to scene entities rather than the camera makes these hand–entity blocks exactly viewpoint-invariant, enabling direct human-to-robot transfer; the entity’s own pose $\text{REF}T_E$ remains tied to the reference frame, so robustness to novel viewpoints comes from demonstrations spanning many head poses rather than from exact invariance. We also gain a unified, variable-length interface that accommodates scenes with different numbers of objects without architectural changes. We empirically show that ICT is the key enabler of cross-embodiment transfer (Sec. 4.3, 4.4).

3.4 Flow Matching Policy with Dense Auxiliary Objectives

Our policy (Fig. 2) takes the scene state s_t —ICT tokens and an RGB image—and generates a bi-manual action trajectory $\mathbf{a} \in \mathbb{R}^{K \times D_a}$ over a K -step horizon, where each D_a -dim slice concatenates both hands’ 6-DoF poses and binary grasps. We describe the training below.

Flow matching action generation. We formulate action generation as a conditional flow matching [33, 64] problem: we parameterize a velocity field v_θ with a transformer decoder conditioned on s_t , and train it to transport a Gaussian prior sample to the action target. Our primary training loss is:

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} \left[w_p \|\Delta \mathbf{p}\|^2 + w_r \|\Delta \mathbf{r}\|^2 + w_g \|\Delta g\|^2 \right], \quad (2)$$

where w_p, w_r, w_g are the loss weights for position ($\mathbf{p}$), rotation ($\mathbf{r}$), and grasp (g); $\Delta(\cdot) = v_\theta(\mathbf{x}_t, t, s_t) - (\mathbf{x}_1 - \mathbf{x}_0)$ is the velocity prediction error; $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$ is the interpolated sample at flow time $t \sim \mathcal{U}(0, 1)$; $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is a Gaussian prior sample; and $\mathbf{x}_1$ is the ground-truth bimanual action. At inference, we integrate the learned ODE with a fixed-step Euler solver.

Dense auxiliary objectives. To extract rich supervision from every demonstration, we add three auxiliary objectives that share the context encoder with the flow matching head: (1) *Object motion*

($\mathcal{L}_{\text{OM}}$): we predict each manipulated object’s future 6-DoF trajectory, forcing the encoder to model object dynamics under hand motion; (2) *2D trace* ($\mathcal{L}_{\text{2D}}$): we regress future 2D projections of entity trajectories, grounding the representation in the visual observation; (3) *Latent consistency* ($\mathcal{L}_{\text{LC}}$): we predict the ICT state K steps ahead, pushing the encoder to capture scene dynamics. We combine them with the flow matching loss into a single objective:

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_{\text{OM}} \mathcal{L}_{\text{OM}} + \lambda_{\text{2D}} \mathcal{L}_{\text{2D}} + \lambda_{\text{LC}} \mathcal{L}_{\text{LC}}, \quad (3)$$

where λ_{OM} , λ_{2D} , λ_{LC} are the loss weights of the three auxiliary objectives. We derive every auxiliary target automatically from the perception pipeline, so each demonstration yields a dense multi-task signal. All three objectives forecast how the scene evolves in complementary spaces (3D physical, 2D visual, latent space), equipping the shared encoder with a lightweight world model of hand–object interaction. We also exploit the shared encoder as a multi-task regularizer that curbs overfitting, with the largest gains in the low-data regime (Sec. 4.2, 4.4).

3.5 Deployment on a Robot

Nothing inside the policy changes when it moves from human video to a robot: it consumes the same two inputs as in training—one embodiment-agnostic RGB image and a set of ICT tokens—and only the modules that *produce* them are swapped. At deployment the camera is static, so its frame is the reference frame REF directly; on human video we instead freeze the camera pose at the first manipulation frame, so each demonstration contributes its own REF and a new camera placement is another draw from that distribution rather than an out-of-distribution input. The robot-side frontend is the training frontend with triangulation replaced by depth back-projection, needing neither SLAM nor the scene sweep: detection, segmentation, and orientation estimation run *once* per episode from one text prompt per object, while arm forward kinematics supplies T_{ee} and g every cycle. Grasp latching carries over unchanged (Sec. 3.3) with the end effector in place of the hand, so a grasped object keeps a fixed pose relative to the gripper. The loop is synchronous at ~ 10 Hz on a single RTX 4090, so each plan acts on a frame at most ~ 100 ms old. The only per-rig setup is camera intrinsics and one hand–eye extrinsic per arm; we use no CAD models, no manual initialization, and no depth inside the policy. Appendix G.1 details the full pipeline.

4 Experiments

We evaluate HumanEgo to answer four questions: (1) Can the embodiment gap be bridged to achieve reliable manipulation from human video alone? (Sec. 4.1) (2) How does policy performance scale with human data versus matched robot data? (Sec. 4.2) (3) How robust is the policy to distribution shifts in embodiment, viewpoint, and environment? (Sec. 4.3) (4) How much does each component contribute to the final performance? (Sec. 4.4) Unless otherwise noted, all experiments are conducted on Trossen WidowX arms with a top-mounted RealSense D405.

4.1 HumanEgo Bridges the Embodiment Gap Efficiently

We evaluate HumanEgo on four real-world manipulation tasks (Fig. 3): *Serve Bread*, a **pick-and-place** task in which the robot grasps a croissant from arbitrary positions and places it on a plate; *Downstack Cups*, a **long-horizon multi-step** task requiring sequential toppling, grasping, and re-stacking of three nested cups; *Water Flowers*, a **contact-rich bimanual** task with strict temporal ordering—one arm holds a pulled-out spray head over a flower pot while the other opens the valve; and *Adjust Table*, a **sustained rotational-control** task in which the robot grasps a crank handle and turns it three full revolutions without releasing. Every trial starts from a randomized initial object position, and we report success rate (%) over 40 trials per task. We compare against five zero-shot methods that learn manipulation from human egocentric video, spanning three design families: point-based (*EgoZero* [21], *Point Policy* [20]), goal-conditioned (*ZeroMimic* [51], *Track2Act* [19]), and object-centric (*SPOT* [48]); Appendix C describes each. All five are trained on the same 30 minutes of human data per task, are reimplemented in our codebase, and receive the same perception frontend,

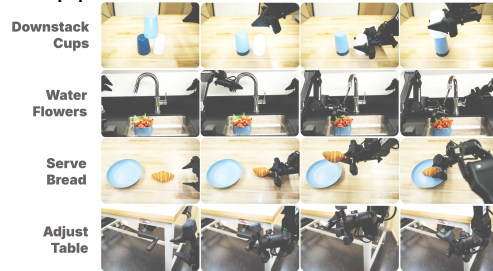


Fig. 3: **Four Real-World Evaluation tasks.**

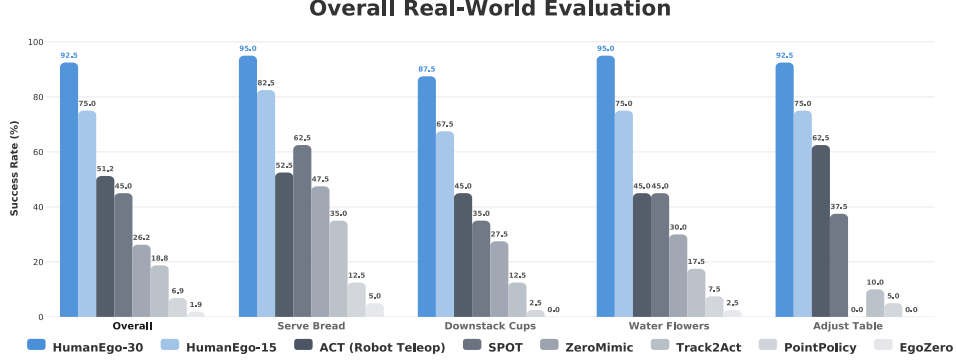


Fig. 4: **Overall Real-World Evaluation.** Real-world success rate (%) for each method across all four tasks. HumanEgo with 30 min of data achieves the highest success rate on every task, demonstrating consistent improvements over both human-video baselines and robot teleoperation methods.

controller, and post-processing as HumanEgo, including the grasp latching of Sec. 3.3, so differences reflect the representation and policy rather than the surrounding system. We also include ACT [28], trained on 30 minutes of robot teleoperation collected on the same hardware. A human demonstration takes about 40 s, beginning with a ~ 1 s scene sweep that supplies the parallax for triangulation, whereas an experienced operator needs about 60 s per teleoperated demonstration.

HumanEgo achieves the highest success rate on every single task. As shown in Fig. 4, HumanEgo with only 30 minutes of human data reaches **92.5%** average success rate across all four tasks. In contrast, the five human-video baselines range from 1.9% to 45.0%: each captures only a partial aspect of manipulation, performing adequately on simpler tasks but collapsing on those that demand precise hand-object reasoning. The lowest scores come from the point-based methods, which encode the scene as unstructured points in absolute coordinates with no entity separation and no relative hand-object transform; their failures concentrate at the grasp, where the gripper arrives offset from the randomized object and the episode cannot recover. HumanEgo is the only method that maintains high performance regardless of task complexity.

Even with half the data, HumanEgo outperforms robot teleoperation. HumanEgo with only 15 minutes of human data already reaches 75.0%, surpassing ACT trained on 30 minutes of robot teleoperation (51.2%), highlighting the data efficiency of our approach—and that *readily available human videos can be a surprisingly potent data source for robot learning*.

HumanEgo excels on tasks that demand precise coordination and spatial reasoning. On Downstack Cups, where ~ 1 cm tolerance makes early errors compound, HumanEgo reaches 87.5% while no baseline exceeds 45%; on Water Flowers, which demands strict bimanual ordering and precise aiming of the stream into the pot, it achieves 95%, more than double the best baseline (45%).

4.2 How Does Performance Scale with Human Data?

We compare HumanEgo trained on human video against ACT and HumanEgo trained on robot teleoperation as a function of collection time on Serve Bread (Fig. 5).

HumanEgo learns effectively from minimal human data. With only ~ 7 minutes of human demonstrations, HumanEgo already reaches 50% success rate and continues to climb smoothly, reaching 95% at 30 minutes. This steep, monotonic scaling curve indicates that our pipeline extracts useful manipulation signal from even a handful of demonstrations.

Auxiliary objectives amplify learning when demonstrations are scarce. Between 2 and

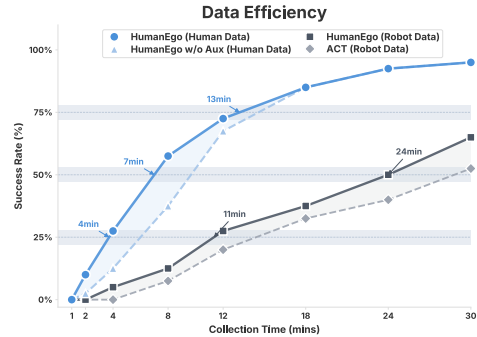


Fig. 5: **Data efficiency.** Success rate (%) vs. data collection time. HumanEgo trained on 8 min of human data surpasses ACT’s 30-min robot data.

12 minutes, HumanEgo with auxiliary losses consistently outperforms the variant without them, with the largest gain at 8 min (57.5% vs. 37.5%). At 18 minutes and beyond the two variants coincide, both reaching 95% at 30 minutes, confirming that auxiliary losses extract richer supervision from each demonstration—a benefit that diminishes as data grows abundant.

Human video is a more efficient data source than robot teleoperation. At 8 minutes of collection time, HumanEgo trained on human video (57.5%) already surpasses ACT trained on 30 minutes of robot teleoperation (52.5% on this task)—a $3.75\times$ reduction in collection effort. Holding the architecture fixed isolates the data source: the same policy trained on 30 min of robot teleoperation reaches 65% versus 95% on human video (+30 pp), so the gain is not an artifact of the policy class.

Diversity, not quantity alone, drives the gap. Offline metrics (Fig. 6) show where the extra signal comes from: per minute of collection, human demonstrations have higher signal-to-noise ratio, smoother motion, and less idle time, and they spread more widely over object positions, trajectory shapes, and sessions. Matched minutes of teleoperation therefore buy fewer and less varied interactions.

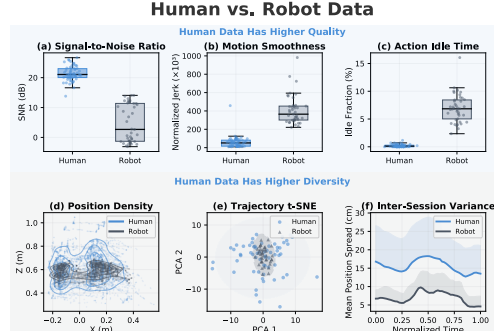


Fig. 6: **Human vs. robot data.** Human egocentric data exhibits higher SNR, smoother motion, less idle time (top), and greater spatial and trajectory diversity (bottom).

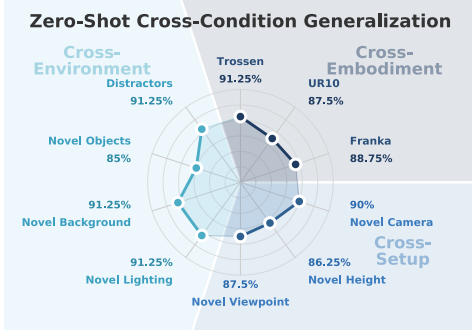


Fig. 7: **Zero-Shot Cross-Condition Generalization.** HumanEgo maintains robust success across different conditions without retraining.



Fig. 8: **Cross-condition real-world evaluation:** cross-embodiment / environment / setup.

4.3 One Policy, Many Conditions

We deploy HumanEgo on Serve Bread and Downstack Cups, each trained on 30 minutes of human data, across the 9 out-of-distribution conditions of Fig. 7—Trossen is the in-distribution reference—with no retraining, and report the mean success rate over the two tasks (setup in Fig. 8).

HumanEgo is robust across the tested visual conditions. Changing the background, lighting, or viewpoint, adding distractors, or swapping in object instances that never appear in the training set all yield 85–91.25% success, with no measurable degradation in most cases. Each condition varies one factor at a time, so this demonstrates that the policy extracts task-relevant information from the visual input while remaining insensitive to the nuisance factors we tested.

HumanEgo is robust to varied object placements. On Serve Bread, it delivers the bread to the plate across a wide range of absolute and relative positions on the table and even at novel heights; on Downstack Cups, it completes the three-step sequence under varied table heights and cup positions—scenarios where methods like EgoZero [21] and Point Policy [20] often fail. This robustness reflects the object-centric structure of our interaction-centric representation.

HumanEgo is hardware-agnostic. All training data is collected with Aria glasses, yet the same checkpoint drives a Trossen, Franka, or UR10 arm under a different camera, sharing no hardware

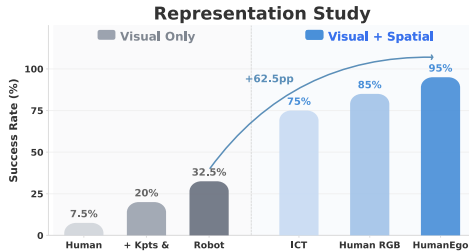


Fig. 9: **Representation study.** Success rate (%) for six input configurations. Visual-only methods plateau at 32.5% with any strategy, whereas the full system reaches 95% (+62.5 pp).

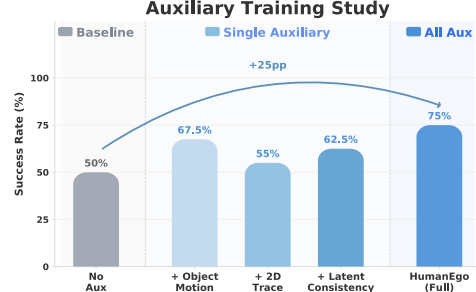


Fig. 10: **Auxiliary training study.** Success rate (%) at 15 min of data for each auxiliary objective individually. Each improves over the baseline; all three combined give +25 pp.

with the training setup. It transfers once each arm’s hand-eye extrinsic is known (Sec. 3.5)—enabled by the interaction-centric representation detailed in Sec. 3.3.

4.4 What Drives Performance of HumanEgo?

We ablate HumanEgo’s two core design choices on Water Flowers: the spatial representation at 30 minutes of data (Fig. 9) and the auxiliary training objectives at 15 minutes (Fig. 10).

Explicit spatial representation, not visual fidelity alone, is the key to bridging the embodiment gap. We isolate visual preprocessing from spatial representation. Progressively reducing the visual embodiment gap—from raw human RGB (7.5%) to keypoint rendering with arm inpainting (20%) to robot RGB that eliminates the gap entirely (32.5%)—yields only modest gains; even with zero visual mismatch, the policy barely exceeds 30%. Monocular RGB encodes *appearance*, not the 3D spatial relationships that manipulation demands. Adding ICT to raw human RGB produces a dramatic jump from 7.5% to 85%, and the full system reaches 95%. In the other direction, removing the image branch so that visual information reaches the policy only through ICT yields 75%: the image stream contributes 20 pp on top of ICT. Since ICT is itself derived from vision, this variant routes visual information through entities rather than removing it. ICT directly encodes the relative 6-DoF transforms between hands and objects—the core manipulation state—turning the problem of inferring 3D dynamics from pixels into learning actions from explicit spatial relationships.

Auxiliary objectives provide complementary gains. We evaluate each objective individually at 15 minutes of data. Each one improves over the no-auxiliary baseline—object motion (+17.5 pp), 2D trace (+5 pp), and latent consistency (+12.5 pp)—and combining all three yields +25 pp. Together, the ablations support the main experimental thesis: ICT supplies the transferable manipulation state, and dense auxiliary supervision makes that state learnable from minutes of human video.

5 Conclusion

We presented **HumanEgo**, a framework that learns robot manipulation policies from minutes of human egocentric video—without any robot data or large-scale pretraining. A hardware-agnostic representation bridges the embodiment gap through visual preprocessing (arm inpainting, keypoint rendering) and spatial encoding (ICT), while a flow matching policy with dense auxiliary objectives learns from scarce data. HumanEgo reaches 92.5% average success across four real-world tasks, improves on matched-time robot teleoperation by 41 percentage points, and transfers zero-shot to novel embodiments, cameras, and environments without retraining.

Limitations and future work. HumanEgo relies on Aria’s stereo hand tracking; monocular substitutes drop real-world success sharply. Object poses are estimated once per episode and then propagated kinematically through the hand—or the gripper at deployment—so we assume a quasi-static scene: in-hand reorientation, slip, or re-grasping violates it. Retargeting maps the thumb-index pair onto a parallel-jaw pose and a binary grasp, so transfer is demonstrated across parallel-jaw arms rather than dexterous hands. Off-the-shelf perception failures also cascade into the policy, and few-shot learning plateaus at ~ 1 cm precision. Appendix B expands these and outlines future work.

References

- [1] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Choromanski, T. Ding, D. Driess, A. Dubey, C. Finn, P. Florence, C. Fu, M. G. Arenas, K. Gopalakrishnan, K. Han, K. Hausman, A. Herzog, J. Hsu, B. Ichter, A. Irpan, N. Joshi, R. Julian, D. Kalashnikov, Y. Kuang, I. Leal, L. Lee, T.-W. E. Lee, S. Levine, Y. Lu, H. Michalewski, I. Mordatch, K. Pertsch, K. Rao, K. Reymann, M. Ryoo, G. Salazar, P. Sanketi, P. Sermanet, J. Singh, A. Singh, R. Soricut, H. Tran, V. Vanhoucke, Q. Vuong, A. Wahid, S. Welker, P. Wohlhart, J. Wu, F. Xia, T. Xiao, P. Xu, S. Xu, T. Yu, and B. Zitkovich. RT-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.
- [2] ALOHA 2 Team, J. Aldaco, T. Armstrong, R. Baruch, J. Bingham, S. Chan, K. Draper, D. Dwibedi, C. Finn, P. Florence, S. Goodrich, W. Gramlich, T. Hage, A. Herzog, J. Hoech, T. Nguyen, I. Storz, B. Tabanpour, L. Takayama, J. Thompson, A. Wahid, T. Wahrburg, S. Xu, S. Yaroshenko, K. Zakka, and T. Z. Zhao. ALOHA 2: An enhanced low-cost hardware for bimanual teleoperation. *arXiv preprint arXiv:2405.02292*, 2024.
- [3] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *Robotics: Science and Systems (RSS)*, 2023.
- [4] A. Khazatsky, K. Pertsch, S. Nair, A. Balakrishna, S. Dasari, S. Karamcheti, S. Nasiriany, M. K. Srirama, L. Y. Chen, K. Ellis, P. D. Fagan, J. Hejna, M. Itkina, M. Lepert, Y. J. Ma, P. T. Miller, J. Wu, S. Belkhale, S. Dass, H. Ha, A. Jain, A. Lee, Y. Lee, M. Memmel, S. Park, I. Radosavovic, K. Wang, A. Zhan, K. Black, C. Chi, K. B. Hatch, S. Lin, J. Lu, J. Mercat, A. Rehman, P. R. Sanketi, A. Sharma, C. Simpson, Q. Vuong, H. R. Walke, B. Wulfe, T. Xiao, J. H. Yang, A. Yavary, T. Z. Zhao, C. Agia, R. Baijal, M. G. Castro, D. Chen, Q. Chen, T. Chung, J. Drake, E. P. Foster, J. Gao, V. Guizilini, D. A. Herrera, M. Heo, K. Hsu, J. Hu, M. Z. Irshad, D. Jackson, C. Le, Y. Li, K. Lin, R. Lin, Z. Ma, A. Maddukuri, S. Mirchandani, D. Morton, T. Nguyen, A. O’Neill, R. Scalise, D. Seale, V. Son, S. Tian, E. Tran, A. E. Wang, Y. Wu, A. Xie, J. Yang, P. Yin, Y. Zhang, O. Bastani, G. Berseth, J. Bohg, K. Goldberg, A. Gupta, A. Gupta, D. Jayaraman, J. J. Lim, J. Malik, R. Martín-Martín, S. Ramamoorthy, D. Sadigh, S. Song, J. Wu, M. C. Yip, Y. Zhu, T. Kollar, S. Levine, and C. Finn. DROID: A large-scale in-the-wild robot manipulation dataset. In *Robotics: Science and Systems (RSS)*, 2024.
- [5] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. P. Foster, P. R. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn. OpenVLA: An open-source vision-language-action model. In *8th Annual Conference on Robot Learning*, 2024. URL <https://openreview.net/forum?id=ZMnD6QZAE6>.
- [6] K. Black, N. Brown, J. Darpinian, K. Dhahalia, D. Driess, A. Esmail, M. R. Equi, C. Finn, N. Fusai, M. Y. Galliker, D. Ghosh, L. Groom, K. Hausman, brian ichter, S. Jakubczak, T. Jones, L. Ke, D. LeBlanc, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, J. Tanner, Q. Vuong, H. Walke, A. Walling, H. Wang, L. Yu, and U. Zhilinsky. $\pi_{0.5}$: a vision-language-action model with open-world generalization. In *9th Annual Conference on Robot Learning*, 2025. URL <https://openreview.net/forum?id=vlhoswksB0>.
- [7] J. Engel, K. Somasundaram, M. Goesele, A. Sun, A. Gamino, A. Turner, A. Talattof, A. Yuan, B. Souti, B. Meredith, C. Peng, C. Sweeney, C. Wilson, D. Barnes, D. DeTone, D. Caruso, D. Valleroy, D. Ginjupalli, D. Frost, E. Miller, E. Mueggler, E. Oleinik, F. Zhang, G. Somasundaram, G. Solaira, H. Lanaras, H. Howard-Jenkins, H. Tang, H. J. Kim, J. Rivera, J. Luo, J. Dong, J. Straub, K. Bailey, K. Eickenhoff, L. Ma, L. Pesqueira, M. Schwesinger, M. Monge, N. Yang, N. Charron, N. Raina, O. Parkhi, P. Borschowa, P. Moulon, P. Gupta, R. Mur-Artal, R. Pennington, S. Kulkarni, S. Miglani, S. Gondi, S. Solanki, S. Diener, S. Cheng, S. Green, S. Saarinen, S. Patra, T. Mourikis, T. Whelan, T. Singh, V. Balntas, V. Baiyya, W. Dreeves, X. Pan, Y. Lou, Y. Zhao, Y. Mansour, Y. Zou, Z. Lv, Z. Wang, M. Yan, C. Ren, R. De Nardi,

- and R. Newcombe. Project Aria: A new tool for egocentric multi-modal AI research. *arXiv preprint arXiv:2308.13561*, 2023.
- [8] S. Kareer, D. Patel, R. Punamiya, P. Mathur, S. Cheng, C. Wang, J. Hoffman, and D. Xu. EgoMimic: Scaling imitation learning via egocentric video. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
 - [9] R. Punamiya, D. Patel, P. Aphiwetsa, P. Kuppili, L. Y. Zhu, S. Kareer, J. Hoffman, and D. Xu. EgoBridge: Domain adaptation for generalizable imitation from egocentric human data. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
 - [10] Y. Liu, W. C. Shin, Y. Han, Z. Chen, H. Ravichandar, and D. Xu. ImMimic: Cross-domain imitation from human videos via mapping and interpolation. In *Conference on Robot Learning (CoRL)*, 2025.
 - [11] R.-Z. Qiu, S. Yang, X. Cheng, C. Chawla, J. Li, T. He, G. Yan, D. J. Yoon, R. Hoque, L. Paulsen, G. Yang, J. Zhang, S. Yi, G. Shi, and X. Wang. Humanoid policy ~ human policy. In *9th Annual Conference on Robot Learning*, 2025. URL <https://openreview.net/forum?id=Tx54fkQ3Cq>.
 - [12] R. Yang, Q. Yu, Y. Wu, R. Yan, B. Li, A.-C. Cheng, X. Zou, Y. Fang, X. Cheng, R.-Z. Qiu, H. Yin, S. Liu, S. Han, Y. Lu, and X. Wang. EgoVLA: Learning vision-language-action models from egocentric human videos. *arXiv preprint arXiv:2507.12440*, 2025.
 - [13] R. Zheng, D. Niu, Y. Xie, J. Wang, M. Xu, Y. Jiang, F. Castañeda, F. Hu, Y. L. Tan, L. Fu, T. Darrell, F. Huang, Y. Zhu, D. Xu, and L. Fan. EgoScale: Scaling dexterous manipulation with diverse egocentric human data. *arXiv preprint arXiv:2602.16710*, 2026.
 - [14] R. Punamiya, S. Kareer, Z. Liu, J. Citron, R.-Z. Qiu, X. Cai, A. Gavryushin, J. Chen, D. Liconti, L. Y. Zhu, P. Aphiwetsa, B. Li, A. Cheluva, P. Kuppili, Y. Liu, D. Patel, M. Pollefeys, R. Katzschmann, X. Wang, S. Song, J. Hoffman, D. Xu, et al. EgoVerse: An egocentric human dataset for robot learning from around the world. *arXiv preprint arXiv:2604.07607*, 2026.
 - [15] R. Hoque, P. Huang, D. J. Yoon, M. Sivapurapu, and J. Zhang. EgoDex: Learning dexterous manipulation from large-scale egocentric video. In *International Conference on Learning Representations (ICLR)*, 2026.
 - [16] M. Lepert, J. Fang, and J. Bohg. Phantom: Training robots without robots using only human videos. In *Conference on Robot Learning (CoRL)*, 2025.
 - [17] M. Lepert, J. Fang, and J. Bohg. Masquerade: Learning from in-the-wild human videos using data-editing. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
 - [18] E. Dessalene, P. Mantripragada, M. Maynord, and Y. Aloimonos. EmbodiSwap for zero-shot robot imitation learning. *arXiv preprint arXiv:2510.03706*, 2025.
 - [19] H. Bharadhwaj, R. Mottaghi, A. Gupta, and S. Tulsiani. Track2Act: Predicting point tracks from internet videos enables generalizable robot manipulation. In *European Conference on Computer Vision (ECCV)*, 2024.
 - [20] S. Haldar and L. Pinto. Point policy: Unifying observations and actions with key points for robot manipulation. In *Conference on Robot Learning (CoRL)*, 2025.
 - [21] V. Liu, A. Adeniji, H. Zhan, S. Haldar, R. Bhirangi, P. Abbeel, and L. Pinto. EgoZero: Robot learning from smart glasses. *arXiv preprint arXiv:2505.20290*, 2025.
 - [22] C. Wang, L. Fan, J. Sun, R. Zhang, L. Fei-Fei, D. Xu, Y. Zhu, and A. Anandkumar. MimicPlay: Long-horizon imitation learning by watching human play. In *Conference on Robot Learning (CoRL)*, 2023.

- [23] G. Li, Y. Lyu, Z. Liu, C. Hou, J. Zhang, and S. Zhang. H2R: A human-to-robot data augmentation for robot pre-training from videos. *arXiv preprint arXiv:2505.11920*, 2025.
- [24] M. Xu, Z. Xu, C. Chi, M. Veloso, and S. Song. XSkill: Cross embodiment skill discovery. In *7th Annual Conference on Robot Learning*, 2023. URL <https://openreview.net/forum?id=8L6pHd9aS6w>.
- [25] M. Xu, Z. Xu, Y. Xu, C. Chi, G. Wetzstein, M. Veloso, and S. Song. Flow as the cross-domain manipulation interface. In *Conference on Robot Learning (CoRL)*, 2024.
- [26] V. Jain, M. Attarian, N. J. Joshi, A. Wahid, D. Driess, Q. Vuong, P. R. Sanketi, P. Sermanet, S. Welker, C. Chan, I. Gilitschenski, Y. Bisk, and D. Dwibedi. Vid2Robot: End-to-end video-conditioned policy learning with cross-attention transformers. In *Robotics: Science and Systems (RSS)*, 2024.
- [27] C. Wen, X. Lin, J. So, K. Chen, Q. Dou, Y. Gao, and P. Abbeel. Any-point trajectory modeling for policy learning, 2024. URL <https://arxiv.org/abs/2401.00025>.
- [28] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Robotics: Science and Systems (RSS)*, 2023.
- [29] K. Yu, S. Zhang, H. Soora, F. Huang, H. Huang, P. Tokekar, and R. Gao. GenFlowRL: Shaping rewards with generative object-centric flow in visual reinforcement learning. *arXiv preprint arXiv:2508.11049*, 2025.
- [30] H. Li, L. Sun, Y. Hu, D. Ta, J. Barry, G. Konidaris, and J. Fu. NovaFlow: Zero-shot manipulation via actionable flow from generated videos. *arXiv preprint arXiv:2510.08568*, 2025.
- [31] S. Patel, S. Mohan, H. Mai, U. Jain, S. Lazebnik, and Y. Li. Robotic manipulation by imitating generated videos without physical demonstrations. *arXiv preprint arXiv:2507.00990*, 2025.
- [32] K. Yu, Y. Han, Q. Wang, V. Saxena, D. Xu, and Y. Zhao. Mimictouch: Leveraging multi-modal human tactile demonstrations for contact-rich manipulation. In *8th Annual Conference on Robot Learning*, 2024. URL <https://openreview.net/forum?id=7yMZAUKXa4>.
- [33] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.
- [34] K. Grauman, A. Westbury, E. Byrne, Z. Chavis, A. Furnari, R. Girdhar, J. Hamburger, H. Jiang, M. Liu, X. Liu, M. Martin, T. Nagarajan, I. Radosavovic, S. K. Ramakrishnan, F. Ryan, J. Sharma, M. Wray, M. Xu, E. Z. Xu, C. Zhao, et al. Ego4D: Around the world in 3,000 hours of egocentric video. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [35] D. Damen, H. Doughty, G. M. Farinella, A. Furnari, E. Kazakos, J. Ma, D. Moltisanti, J. Munro, T. Perrett, W. Price, and M. Wray. Rescaling egocentric vision: Collection pipeline and challenges for EPIC-KITCHENS-100. *International Journal of Computer Vision (IJCV)*, 2022.
- [36] P. Banerjee, S. Shkodrani, P. Moulon, S. Hampali, F. Zhang, J. Fountain, E. Miller, S. Basol, R. Newcombe, R. Wang, J. J. Engel, and T. Hodan. Introducing HOT3D: An egocentric dataset for 3d hand and object tracking. *arXiv preprint arXiv:2406.09598*, 2024.
- [37] Y. Liu, Y. Liu, C. Jiang, K. Lyu, W. Wan, H. Shen, B. Liang, Z. Fu, H. Wang, and L. Yi. HOI4D: A 4d egocentric dataset for category-level human-object interaction. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [38] Y. Liu, H. Yang, X. Si, L. Liu, Z. Li, Y. Zhang, Y. Liu, and L. Yi. TACO: Benchmarking generalizable bimanual tool-ACTION-object understanding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

- [39] X. Wang, T. Kwon, M. Rad, B. Pan, I. Chakraborty, S. Andrist, D. Bohus, A. Feniello, B. Tekin, F. V. Frujeri, N. Joshi, and M. Pollefeys. HoloAssist: An egocentric human interaction dataset for interactive AI assistants in the real world. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [40] G. Zhang, Q. Xu, H. Zhang, J. Ma, L. He, Y. Bao, Z. Ping, Z. Yuan, C. Lu, C. Yuan, T. Liang, X. Tian, M. Shao, F. Zhang, M. Ding, Y. Gao, H. Zhao, H. Zhao, and H. Xu. UniDex: A robot foundation suite for universal dexterous hand control from egocentric human videos. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [41] S. Lee, Y. Jung, I. Chun, Y.-C. Lee, Z. Cai, H. Huang, A. Talreja, T. D. Dao, Y. Liang, J.-B. Huang, and F. Huang. TraceGen: World modeling in 3d trace-space enables learning from cross-embodiment videos. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
- [42] C. Yuan, C. Wen, T. Zhang, and Y. Gao. General flow as foundation affordance for scalable robot learning. In *8th Annual Conference on Robot Learning*, 2024. URL <https://openreview.net/forum?id=nmEt0ci8hi>.
- [43] L. Y. Zhu, P. Kuppili, R. Punamiya, P. Aphiwetsa, D. Patel, S. Kareer, S. Ha, and D. Xu. EMMA: Scaling mobile manipulation via egocentric human data. *IEEE Robotics and Automation Letters*, 11(3):3087–3094, 2026.
- [44] S. Kareer, K. Pertsch, J. Darpinian, J. Hoffman, D. Xu, S. Levine, C. Finn, and S. Nair. Emergence of human to robot transfer in vision-language-action models. *arXiv preprint arXiv:2512.22414*, 2025.
- [45] H. Kim, J. Kang, H. Kang, M. Cho, S. J. Kim, and Y. Lee. Uniskill: Imitating human videos via cross-embodiment skill representations. In *9th Annual Conference on Robot Learning*, 2025. URL <https://openreview.net/forum?id=EgSDP6A0F1>.
- [46] I. Guzey, H. Qi, J. Urain, C. Wang, J. Yin, K. Bodduluri, M. Lambeta, L. Pinto, A. Rai, J. Malik, T. Wu, A. Sharma, and H. Bharadhwaj. Dexterity from smart lenses: Multi-fingered robot manipulation with in-the-wild human demonstrations. *arXiv preprint arXiv:2511.16661*, 2025.
- [47] A. Singh, K. Torshizi, K. Habib, K. Yu, R. Gao, and P. Tokekar. Afford2Act: Affordance-guided automatic keypoint selection for generalizable and lightweight robotic manipulation. *arXiv preprint arXiv:2510.01433*, 2025.
- [48] C.-C. Hsu, B. Wen, J. Xu, Y. Narang, X. Wang, Y. Zhu, J. Biswas, and S. Birchfield. SPOT: SE(3) pose trajectory diffusion for object-centric manipulation. *arXiv preprint arXiv:2411.00965*, 2024.
- [49] Y. Zou, C. Shi, W. Yu, H. Xue, J. Lv, Y. Pan, C. Wen, and C. Lu. ActiveGlasses: Learning manipulation with active vision from ego-centric human demonstration. *arXiv preprint arXiv:2604.08534*, 2026.
- [50] Z.-H. Yin, S. Yang, and P. Abbeel. Object-centric 3d motion field for robot learning from human videos. *arXiv preprint arXiv:2506.04227*, 2025.
- [51] J. Shi, Z. Zhao, T. Wang, I. Pedroza, A. Luo, J. Wang, J. Ma, and D. Jayaraman. ZeroMimic: Distilling robotic manipulation skills from web videos. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [52] S. Park, H. Bharadhwaj, and S. Tulsiani. DemoDiffusion: One-shot human imitation using pre-trained diffusion policy. *arXiv preprint arXiv:2506.20668*, 2025.
- [53] R. Shah, S. Liu, Q. Wang, Z. Jiang, S. Kumar, M. Seo, R. Martín-Martín, and Y. Zhu. MimicDroid: In-context learning for humanoid robot manipulation from human play videos. *arXiv preprint arXiv:2509.09769*, 2025.

- [54] H. Chen, T. Dong, T. Wu, L. Wang, Y. Jangir, Y. Niu, Y. Ye, H. Bharadhwaj, Z. Erickson, and J. Ichnowski. Dexterous manipulation policies from RGB human videos via 3d hand-object trajectory reconstruction. *arXiv preprint arXiv:2602.09013*, 2026.
- [55] J. Shi, J. Smith, J. Qian, and D. Jayaraman. Points2Reward: Robotic manipulation rewards from just one video. In *RSS Workshop on Semantic Robotics (SemRob)*, 2025.
- [56] B. Wang, N. Sridhar, C. Feng, M. van der Merwe, A. Fishman, N. Fazeli, and J. J. Park. This&that: Language-gesture controlled video generation for robot planning. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 12842–12849, 2025. doi: [10.1109/ICRA55743.2025.11128780](https://doi.org/10.1109/ICRA55743.2025.11128780).
- [57] H. Xiong, Q. Li, Y.-C. Chen, H. Bharadhwaj, S. Sinha, and A. Garg. Learning by watching: Physical imitation of manipulation skills from human videos. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 7827–7834, 2021. doi:[10.1109/IROS51168.2021.9636080](https://doi.org/10.1109/IROS51168.2021.9636080).
- [58] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár, and C. Feichtenhofer. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.
- [59] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov, N. Kong, H. Goka, K. Park, and V. Lempitsky. Resolution-robust large mask inpainting with Fourier convolutions. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022.
- [60] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, Q. Jiang, C. Li, J. Yang, H. Su, J. Zhu, and L. Zhang. Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection. In *European Conference on Computer Vision (ECCV)*, 2024.
- [61] N. Karaev, I. Rocco, B. Graham, N. Neverova, A. Vedaldi, and C. Rupprecht. CoTracker: It is better to track together. In *European Conference on Computer Vision (ECCV)*, 2024.
- [62] Z. Wang, Z. Zhang, J. Xu, J. Wang, T. Pang, C. Du, H. Zhao, and Z. Zhao. Orient anything V2: Unifying orientation and rotation understanding. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [63] Y. Zhou, C. Barnes, J. Lu, J. Yang, and H. Li. On the continuity of rotation representations in neural networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [64] X. Liu, C. Gong, and Q. Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *International Conference on Learning Representations (ICLR)*, 2023.
- [65] R. A. Potamias, J. Zhang, J. Deng, and S. Zafeiriou. WiLoR: End-to-end 3d hand localization and reconstruction in-the-wild. *arXiv preprint arXiv:2409.12259*, 2024.
- [66] G. Pavlakos, D. Shan, I. Radosavovic, A. Kanazawa, D. Fouhey, and J. Malik. Reconstructing hands in 3D with transformers. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [67] Z. Yu, S. Zafeiriou, and T. Birdal. Dyn-HaMR: Recovering 4d interacting hand motion from a dynamic camera. *arXiv preprint arXiv:2412.12861*, 2025.
- [68] J. Zhang, J. Deng, C. Ma, and R. A. Potamias. HaWoR: World-space hand motion reconstruction from egocentric videos. *arXiv preprint arXiv:2501.02973*, 2025.

- [69] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. G. Yong, J. Lee, W.-T. Chang, W. Hua, M. Georg, and M. Grundmann. MediaPipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*, 2019.
- [70] X. Zhang, Z. Kou, C. Qin, M. Huang, E. Ristani, A. Kumar Lele, L. Chen, K. He, A. Boularias, and L. Guan. Glove2Hand: Synthesizing natural hand-object interaction from multi-modal sensing gloves. *arXiv preprint arXiv:2603.20850*, 2026.
- [71] A. Sarker, Z. Kou, E. Ristani, L. Guan, and T. Niehues. Real-time hand pose tracking using 6-axis IMUs. In *ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 2026.

Appendix

A Extended Related Work

Recent years have produced rich large-scale egocentric and hand–object interaction datasets [14, 15, 34–39] that provide the data foundation for learning manipulation from human video. Building on this foundation, one line of work scales up generalist policies and world models [12, 13, 40–42] that learn embodiment-agnostic representations from massive corpora, yet deployment demands enormous compute and per-task robot post-training. Another line co-trains [8–11, 26, 43, 44] on paired human and robot data, jointly optimizing across embodiments to amortize the human signal, yet every new task still requires its own batch of robot demonstrations. Visual retargeting approaches [16–18, 23] synthesize pseudo-robot demonstrations by inpainting the human arm and rendering a robot in its place, but the rendered imagery is brittle to morphological and viewpoint variations. Hierarchical methods [22, 24, 45] learn high-level plans from human video and delegate low-level control to a robot-trained controller, which still depends on robot data for the low-level skill. To avoid robot data altogether, a third family pursues embodiment-agnostic representations for zero-shot transfer, differing in *what* they represent: point-based methods [20, 21, 46, 47] encode the scene as 2D or 3D points and gain computational efficiency, but lose the structural relationship between hand and object; object-centric methods [29, 31, 48–50] represent the scene through the object’s 6-DoF pose or motion, capturing object dynamics yet modeling the manipulator only implicitly; and goal-conditioned methods [19, 51] predict 2D tracks or 3D wrist trajectories conditioned on a target image, but require explicit goal specification at deployment. Several other directions [25, 30, 52–57] also explore learning manipulation from video along complementary axes. A common thread runs through these zero-shot lines: they represent the hand *or* the object, but rarely their *interaction*—the very signal that defines manipulation. HumanEgo bridges this gap with an interaction-centric representation that explicitly encodes the spatial relationship between hands and objects, achieving zero-shot transfer from only minutes of human egocentric video without any robot data or large-scale pretraining.

B Extended Limitations and Future Work

Limitations and future work. Our framework relies on Aria’s stereo hand tracking—monocular substitutes drop real-world success sharply, calling for stronger monocular hand pose estimators that recover absolute depth. Object poses are estimated once at the start of an episode and thereafter tracked kinematically through the gripper, so we assume a quasi-static scene: unattended objects do not move on their own, and a grasped object does not shift in the hand. In-hand reorientation, slip, or re-grasping violates this assumption and would require online, occlusion-robust tracking. The pipeline chains several off-the-shelf perception modules whose failures cascade into the policy, motivating stronger or jointly-trained frontends. Finally, few-shot learning plateaus at ~ 1 cm precision; reaching sub-centimeter accuracy on contact-rich tasks will likely require reinforcement-learning refinement or simulation-based fine-tuning. We see HumanEgo as a starting point—a fully zero-shot, robot-data-free pipeline that future work can extend with online tracking, learned frontends, and downstream RL refinement.

C Baseline Implementations

The five zero-shot human-video baselines in Sec. 4.1 span three design families. Two are point-based: *EgoZero* [21] lifts the egocentric scene into 3D point clouds that treat hand and object alike, while *Point Policy* [20] instead tracks sparse keypoints on each. Another two are goal-conditioned: *ZeroMimic* [51] distills 3D wrist trajectories from web video and conditions on a goal image at test time, and *Track2Act* [19] predicts 2D point tracks without explicit 3D reasoning. The last, *SPOT* [48], is object-centric, generating SE(3) object trajectories with a diffusion model.

D Data Collection Details

D.1 Aria Gen1 Glasses

Aria Gen1 recording configuration. We record every human demonstration with Project Aria Gen1 glasses (Fig. 11), configured through the official Project Aria Mobile App with the sensor profile listed below:

- **RGB:** 30 fps at 2 MP.
- **SLAM:** 2× monochrome cameras, 30 fps at VGA.
- **ET (eye tracking):** 2× cameras, 10 fps at QVGA.
- **IMUs:** two 6-axis IMUs sampled at 1000 Hz and 800 Hz.
- **Magnetometer, barometer, 7-mic audio array, GPS, Wi-Fi, and BLE:** all enabled for synchronization metadata and environmental context.

All streams are hardware-timestamped on-device and synchronized to a common Aria clock, so every modality is time-aligned at the millisecond level.

Aria Machine Perception Services. On top of the raw recordings, Aria’s cloud-hosted *Machine Perception Services* (MPS) [7] post-process each capture into metric, ready-to-use signals. Two MPS outputs are critical for our pipeline:

- *Closed-loop trajectory.* The closed-loop SLAM output fuses the two monochrome SLAM cameras and the two IMUs and applies loop closure plus global optimization to yield a globally consistent, drift-corrected 6-DoF device trajectory in a gravity-aligned world frame. We query this trajectory at every RGB frame to obtain the calibrated camera extrinsics used for triangulation (App. E.1) and to lift every hand keypoint into the world frame consumed by ICT (Sec. 3.3).
- *Hand tracking.* The MPS hand tracker jointly processes the stereo SLAM cameras to produce 21 3D keypoints per hand (five per finger plus the wrist), reported directly in the world frame with per-keypoint confidence scores. This supplies the 3D hand skeleton consumed by our hand-to-gripper retargeting (App. E.3).



Fig. 11: **Data collection setup.**

Together, the closed-loop trajectory and the hand tracker provide a metric 6-DoF camera pose and a 3D hand skeleton at every frame, without any calibration or scene instrumentation beyond wearing the glasses.

D.2 Task Details

We evaluate HumanEgo on four real-world manipulation tasks spanning pick-and-place, multi-step bimanual coordination, contact-rich reasoning, and sustained rotational control (Fig. 3). For each task we describe the scene, per-trial randomization, target behavior, and success/failure criteria used to score the 40 trials per condition reported in the main paper.

Serve Bread. *Scene.* A croissant and a dinner plate sit on a tabletop, plate on the left and bread on the right. *Randomization.* Across trials we independently randomize (i) the horizontal offset between the two objects, sampled from $[0, 50]$ cm, and (ii) their depths along the table’s front-back axis, so the pair is generally *not* colinear. The right arm starts from an arbitrary, non-aligned pose above the bread. *Target behavior.* Grasp the croissant from above, lift it clear of the table, transport it to a release pose above the center of the plate, and open the gripper. *Success.* The croissant comes to rest on the plate, in any orientation. *Failure.* (a) the bread ends up outside the plate (on the table, draped over the rim, or knocked off); or (b) the bread slips from the gripper during transport.

Downstack Cups. *Scene.* Three cups of distinct colors on a tabletop: a *white* cup at bottom-right, a *dark-blue* cup at bottom-left, and a *light-blue* cup stacked on top of the white cup. The table height and the absolute position of the cup group are varied across trials. *Randomization.* The horizontal gap between the white and dark-blue cups is drawn from $[0, 1]$ cm; the light-blue cup is jittered left/right within $[0, 2]$ cm on top of the white cup. *Target behavior.* A three-step sequence: (1) *Topple*—knock the light-blue cup sideways so it lands on top of the dark-blue cup; (2) *Grasp*—swing over to the white cup and grasp it from above; (3) *Cover*—lower the white cup onto the dark-blue/light-blue stack and release, forming a three-cup tower on the left (dark-blue / light-blue / white, bottom-to-top). *Success.* The three cups end in the intended stable stack on the left. *Failure.* (a) the light-blue cup is never

contacted; (b) it is toppled but does not land on the dark-blue cup; (c) the white cup is not grasped; or (d) the white cup is not correctly placed on the stack (misses it, topples it, or lands off-axis so the tower collapses). Early errors compound, so the policy must succeed at every sub-stage.

Water Flowers. *Scene.* A wall-mounted faucet stands at the front of the workspace, with a sunken sink recessed ~ 10 cm below the tabletop directly underneath. A flower pot filled with fresh flowers sits inside the sink. *Randomization.* The pot is placed at one of three qualitatively different positions in the sink (top-left, middle, or bottom-right); the faucet is free to rotate about its vertical axis within $[-15^\circ, +15^\circ]$. *Target behavior.* Coordinated bimanual execution with strict temporal ordering. **Left arm (spray head):** grasp the pull-out spray head, pull it ~ 15 cm downward out of the faucet socket, and hold the nozzle 3–5 cm above the center of the flower pot, pointed downward. **Right arm (handle):** remain visible in a stationary pre-grasp pose near the faucet handle while the left arm works; once the left arm is in place, grasp the handle and flick it to the right by 3–5 cm to open the valve, so water flows onto the flowers. *Success.* The left arm holds the spray head over the pot while the right arm has opened the valve and water pours onto the flowers. *Failure.* *Left arm:* (a) the spray head is not grasped or not pulled down; (b) it slips during pull-out or transport; or (c) it is not positioned directly above the pot. *Right arm:* (a) the handle is not grasped; or (b) it is not flicked far enough to open the valve, or slips before water flows.

Adjust Table. *Scene.* The operator faces an adjustable table whose height is controlled by a hand crank protruding from its side, oriented roughly horizontally. *Randomization.* The initial angle of the crank handle about its rotation axis is jittered by $\pm 10^\circ$ around horizontal. *Target behavior.* The right arm (1) approaches the crank handle from an arbitrary initial pose and grasps it firmly, then (2) performs a continuous *counter-clockwise* rotation about the crank axis, completing three full revolutions ($3 \times 360^\circ$) without releasing the handle. *Success.* All three revolutions are completed while the grasp is maintained throughout. *Failure.* (a) the handle is never grasped; or (b) the handle slips from the gripper during the rotation, before three full revolutions are completed.

Per-demonstration collection time. Across all four tasks, a single human egocentric demonstration takes approximately **30–40 seconds**, while a matched teleoperated demonstration on the same hardware takes **60–70 seconds**. The roughly $2\times$ gap reflects that everyday human manipulation is naturally faster and more dexterous than teleoperation through a piloting interface; combined with the embodiment-agnostic representation, this makes human video a substantially more sample-efficient training source per minute of collection.

Additional tasks (demonstration-only). Beyond the four tasks evaluated quantitatively above, we also collected data and trained HumanEgo policies on five additional tasks, illustrating the speed with which our pipeline can be brought to bear on new behaviors. These tasks appear in our supplementary demonstration videos but are *not* part of the quantitative evaluation in Sec. 4.1.

Charge Devices. The operator places a smartwatch, a pair of earphones, and a phone from the tabletop onto their corresponding magnetic chargers. Each item is treated as a separate sub-task, following a GRASP $\rightarrow$ TRANSPORT $\rightarrow$ RELEASE-ON-PAD template; we collect and train a dedicated model for each sub-task.

Unscrew Cap. A bimanual task: one hand grasps and stabilizes the bottle body while the other hand grasps the cap, rotates it through 2–3 full revolutions to disengage the thread, and lifts the freed cap aside.

Open Door. The operator opens a self-closing door fitted with a lever-style handle. The acting hand grasps the lever, rotates it clockwise to release the latch, and then pulls the door outward against its return spring.

Open Cabinet. A bimanual task: both hands simultaneously grasp a pair of vertically oriented handles on a cabinet door and pull it open along an outward arc.

Grab Tissue. The operator pinches a single sheet of tissue from a tissue box, lifts it upward and then translates it laterally to fully extract it, and places it on the table.

E Preprocessing Details

E.1 Triangulation

Aria Gen1 glasses lack a depth sensor, so we recover each object’s 3D position by triangulating tracked 2D keypoints across frames, treating the moving head-mounted camera as a multi-view system with calibrated extrinsics given by the 6-DoF Aria MPS SLAM pose [7]. This requires the object to remain stationary during the observation window; once manipulation begins, the object is free to move.

Pre-episode scene sweep. Multi-view triangulation requires the same 3D point to be seen from sufficiently different viewpoints, yet during manipulation the head-mounted camera is often nearly stationary while only the hands move, collapsing the effective camera baseline. We therefore prefix every demonstration with a short *scene sweep*: the demonstrator keeps the scene static and slowly moves their head for ~ 1 – 2 seconds (~ 30 – 60 frames), using either a horizontal left-to-right pan or a forward walk-in toward the object, before proceeding to the actual manipulation.

Multi-view triangulation from 2D tracks. For each object we detect it in the first sweep frame with Grounding DINO [60], segment it with SAM2 [58], sample N keypoints on the resulting mask, and track them through the F sweep frames with CoTracker3 [61]. Let $K \in \mathbb{R}^{3 \times 3}$ be the RGB intrinsics, $T_i = [R_i \mid t_i]$ the camera-to-world SLAM pose of frame i , and $P_i = K [R_i^\top \mid -R_i^\top t_i] \in \mathbb{R}^{3 \times 4}$ the corresponding world-to-image projection. A track $\{\mathbf{u}_n^{(i)}\}_{i=1}^F$ and the unknown 3D point $\mathbf{X}_n \in \mathbb{R}^4$ (homogeneous) satisfy $\mathbf{u}_n^{(i)} \times P_i \mathbf{X}_n = 0$, which yields two linear equations per frame:

$$\begin{bmatrix} u_n^{(i)} \mathbf{p}_3^{(i)\top} - \mathbf{p}_1^{(i)\top} \\ v_n^{(i)} \mathbf{p}_3^{(i)\top} - \mathbf{p}_2^{(i)\top} \end{bmatrix} \mathbf{X}_n = \mathbf{0}, \quad (4)$$

where $\mathbf{p}_j^{(i)\top}$ is the j -th row of P_i . Stacking (4) across all F frames gives a $2F \times 4$ system $A_n \mathbf{X}_n = \mathbf{0}$; we solve it in the least-squares sense via SVD by taking the right singular vector of A_n with the smallest singular value and dehomogenizing to recover $\mathbf{x}_n \in \mathbb{R}^3$. The object position is then the centroid $\mathbf{p}_{\text{obj}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$, which cancels per-point triangulation noise.

E.2 Phase Detection

A raw Aria recording interleaves active manipulation with non-manipulation segments—walking up to the workspace, the pre-episode scene sweep (App. E.1), and stepping back once the task ends. Only the manipulation portions carry clean hand–object dynamics, so we run an automatic *phase detection* step that segments every recording into kinematic modes and keeps only the manipulation frames for training.

Phase taxonomy. Each frame is assigned one of five modes: (0) MANIP—demonstrator stands still and actively manipulates the scene; (1) FORWARD—linear walking; (2) ROTATE—in-place head/body rotation (*e.g.* the scene sweep); (3) TRANSITION—short buffers between adjacent modes; (4) FINISHED—sustained final hold at the end of the recording.

Segmentation signals and training-data selection. Phases are computed from two streams: the 6-DoF head trajectory from Aria SLAM (body motion) and the 3D hand trajectory from the hand tracker (manipulation motion). A frame enters MANIP when the head linear and angular speeds simultaneously fall below $v_{\text{stop}}=0.03$ m/s and $w_{\text{stop}}=0.15$ rad/s for ≥ 15 consecutive frames; ROTATE requires $\|\boldsymbol{\omega}_{\text{head}}\| > 0.10$ rad/s with $\|\mathbf{v}_{\text{head}}\| < 0.08$ m/s; FORWARD collects the remaining high-linear-speed frames; TRANSITION fills a 10-frame buffer at every mode change; and FINISHED is declared once the trailing stop lasts for ≥ 30 frames. We additionally refine MANIP with hand kinematics: a candidate frame is demoted to TRANSITION if the average hand speed exceeds 0.15 m/s over a 5-frame window, trimming reaching/retracting motion away from the manipulation core. The training pipeline then keeps only MANIP (0) and FINISHED (4), dropping FORWARD, ROTATE, and TRANSITION, so the scene sweep, navigation, and mode-change buffers never reach the training signal.

E.3 Hand-to-Gripper Transfer

To treat a human egocentric video as robot data, every frame of the demonstration must carry an end-effector target that a parallel-jaw robot can actually execute. The human hand, however, has 21 articulated keypoints and a morphology very different from a 2-finger gripper, so the raw hand pose cannot be passed through directly. We therefore *retarget* the hand into a virtual gripper—a 6-DoF SE(3) pose plus a 1-DoF grasp scalar—derived from a few anatomically stable keypoints after a short motion-optimization pipeline.

Hand keypoint extraction. We start from the 21-keypoint hand skeleton produced by Aria MPS [7], which fuses the stereo SLAM cameras and the on-device IMU to recover every keypoint’s 3D position in the SLAM world frame at each frame. For retargeting we use only five keypoints per hand (Fig. 12): the *wrist*, *thumb MCP*, *thumb tip*, *index MCP*, and *index tip*.

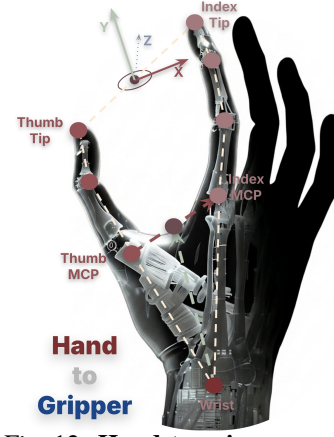


Fig. 12: **Hand-to-gripper mapping.**

Motion optimization. Raw MPS keypoints are noisy and occasionally drop frames, and feeding them directly into the SE(3) construction produces jittery, flip-prone trajectories. We therefore run a short optimization pipeline: (1) *Confidence masking*—we drop any keypoint whose MPS confidence falls below 0.8 and discard detection segments shorter than 30 consecutive frames as likely ghost detections; (2) *Gap interpolation*—short missing intervals (≤ 10 frames) are filled with linear interpolation on positions and SLERP on orientations, so the later smoother sees a dense sequence; (3) *Savitzky–Golay position smoothing*—we apply an SG filter with window size 21 and polynomial order 2 to the five retarget keypoints, removing high-frequency jitter while preserving manipulation-relevant accelerations; (4) *EMA orientation smoothing*—we apply an exponential moving average with $\alpha_x = \alpha_y = 0.15$ to the X - and Y -axes of the gripper frame (defined below), re-orthonormalize via Gram–Schmidt after each update, and enforce sign consistency across adjacent frames to prevent spurious 180° flips.

End-effector position. We take the midpoint of the thumb tip and the index tip as the gripper position, which naturally corresponds to the center of a parallel-jaw grasp:

$$\mathbf{p}_{ee} = \frac{1}{2} (\mathbf{p}_{\text{thumb tip}} + \mathbf{p}_{\text{index tip}}). \quad (5)$$

End-effector orientation. Choosing an orientation that is both *accurate* and *stable through pinch grasps* is the subtle part of retargeting; two natural alternatives both fail. (i) *Raw wrist pose*: using the MPS wrist orientation directly as the gripper frame is inaccurate, because the anatomical wrist frame is not aligned with the thumb–index action axis that the gripper actually uses. (ii) *Wrist-to-fingertip-midpoint*: defining the forward axis as $\text{wrist} \rightarrow \text{mid}(\text{thumb tip}, \text{index tip})$ and the jaw axis as $\text{thumb tip} \rightarrow \text{index tip}$ works when the hand is open but becomes *degenerate at the moment of grasp*—the two fingertips converge to nearly the same point, so the jaw axis collapses to a near-zero vector and the frame is ill-defined. We instead build the gripper frame from the MCP joints, which remain well-separated throughout the full pinch cycle. Writing $\mathbf{p}_w$, $\mathbf{p}_{\text{tMCP}}$, $\mathbf{p}_{\text{iMCP}}$ for the wrist, thumb-MCP, and index-MCP positions, we construct

$$\mathbf{x}_{ee} = \widehat{\mathbf{p}_{\text{iMCP}} - \mathbf{p}_{\text{tMCP}}}, \quad \mathbf{y}_{ee} = \tilde{\mathbf{y}} - (\tilde{\mathbf{y}}^\top \mathbf{x}_{ee}) \mathbf{x}_{ee}, \quad \mathbf{z}_{ee} = \mathbf{x}_{ee} \times \mathbf{y}_{ee}, \quad (6)$$

where $\tilde{\mathbf{y}} = \frac{1}{2} (\mathbf{p}_{\text{tMCP}} + \mathbf{p}_{\text{iMCP}}) - \mathbf{p}_w$ is the raw wrist-to-MCP-midpoint vector and $\widehat{(\cdot)}$ denotes unit normalization. Intuitively, $\mathbf{y}_{ee}$ is the forward axis (wrist $\rightarrow$ MCP midpoint), $\mathbf{x}_{ee}$ is the jaw opening axis (thumb MCP $\rightarrow$ index MCP), and $\mathbf{z}_{ee}$ is the orthogonal complement. The rotation $R_{ee} = [\mathbf{x}_{ee} | \mathbf{y}_{ee} | \mathbf{z}_{ee}] \in \text{SO}(3)$ together with the position $\mathbf{p}_{ee}$ from Eq. (5) gives the SE(3) end-effector pose T_{ee} . Because the two MCPs never collapse to each other during a pinch, this construction stays numerically stable across the full grasp/release cycle.

Gripper aperture. We derive the 1-DoF gripper command from the thumb–index fingertip distance:

$$g = \text{clip} \left(\frac{\|\mathbf{p}_{\text{thumb tip}} - \mathbf{p}_{\text{index tip}}\| - d_{\min}}{d_{\max} - d_{\min}}, 0, 1 \right), \quad (7)$$

where $d_{\min}$ and $d_{\max}$ are the closed and fully-open fingertip distances calibrated per user. The normalized g is then median-filtered and run through a short flicker-suppression pass to produce a clean open/close command stream, and binarized at deployment.

F Training Details

F.1 Flow Matching Policy

Velocity field and loss. We train a conditional flow matching [33] policy that maps a Gaussian prior $\mathbf{x}_0 \sim \mathcal{N}(0, I)$ to the ground-truth bimanual action chunk $\mathbf{x}_1 \in \mathbb{R}^{K \times D_a}$ along the linear path $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$ with flow time $t \sim \mathcal{U}(0, 1)$. The target velocity is the constant displacement $\mathbf{v}_{\text{target}} = \mathbf{x}_1 - \mathbf{x}_0$, and the flow-matching loss is an MSE over the predicted velocity with dimension-wise reweighting: $w_p=5$ on position, $w_r=1$ on 6D rotation, and $w_g=10$ on the grasp logit. We also support an optimal-transport matching variant (OT-CFM) that solves a Hungarian assignment between noise and action samples within each mini-batch before computing the loss, producing straighter target flows; we leave it off by default since we did not find consistent wins on our tasks.

Network. The velocity field v_θ is a 6-layer, 8-head transformer decoder with embedding dimension 384 and dropout 0.05. Each action-chunk token attends (via self-attention) to the rest of the chunk and (via cross-attention) to the conditioning context. Context is built from two streams: (i) the RGB frame, embedded with a 16×16 patch embedding on a 240×320 input and a sinusoidal time embedding fused through a small MLP; and (ii) the state tokens, i.e., the per-entity ICT tokens described in Sec. 3.3, linearly projected to 384 channels.

Auxiliary heads. Three dense auxiliary objectives share the context encoder with the velocity field (Sec. 3.4). The *object motion* head ($\mathcal{L}_{\text{OM}}$) predicts the 9-D future pose trace of the manipulated object and is trained with $0.5(w_p, w_r)$ -weighted MSE; the *2D trace* head ($\mathcal{L}_{2\text{D}}$) emits $K \times 3 \times 2$ normalized image coordinates of three anchor keypoints through a shallow deconvolution stack with loss weight $\lambda_{2\text{D}}=20$; and the *latent consistency* head ($\mathcal{L}_{\text{LC}}$) predicts the hand tokens K steps ahead with a masked MSE weighted by $\lambda_{\text{LC}} \in [0.1, 1.0]$. All three targets are produced automatically by the perception pipeline, so each demonstration yields a dense multi-task signal without extra labeling.

Additional tricks. Two lightweight training tricks further stabilize learning from minutes of data. *Region attention* biases the image cross-attention toward the currently active manipulation anchor: given the anchor’s 2D image projection (u_0, v_0) , we multiply the attention logits by a Gaussian spotlight

$$w(u, v) = \exp\left(-\frac{(u - u_0)^2 + (v - v_0)^2}{2\sigma^2}\right), \quad (8)$$

whose spatial scale σ is a learnable parameter, softly focusing the encoder on task-relevant image regions without hard-cropping. *State-noise injection* perturbs every hand token during training, $\tilde{s}_t = s_t + \epsilon$ with $\epsilon \sim \mathcal{N}(\mathbf{0}, \Sigma_s)$ and separate standard deviations on the position, 6D rotation, and grasp channels, which makes the policy robust to the small perception noise it encounters at deployment.

Optimization recipe. We train with AdamW at a base learning rate of 10^{-4} , cosine decay with 200-step warmup, minimum-LR ratio 0.05, batch size 32, and 400 epochs. We clip gradient norm at 1.0, use bfloat16 mixed precision, and keep an exponential moving average of the weights with decay 0.999 for evaluation and deployment.

Data augmentation. To expand the effective training distribution from only ~ 40 min of human video per task, we apply a cocktail of augmentations on the fly in the dataloader, grouped into three families. (i) *Image augmentations* on the RGB stream. Photometric jitter ($p=0.8$) randomly perturbs brightness (± 0.20), contrast (± 0.20), and gamma (± 0.15), adds Gaussian pixel noise ($\sigma=0.02$), optionally converts the frame to grayscale ($p=0.1$), and jitters HSV hue by ± 10 and saturation by $[0.6, 1.4]$. A random resized crop ($p=0.5$) draws a sub-window with scale in $[0.7, 1.0]$ and aspect ratio in $[0.9, 1.1]$ before resizing back to the network input size. A Gaussian blur with a 3×3 kernel is applied with $p=0.15$, and random erasing ($p=0.5$) overlays 3–8 black cutout patches each covering 5–20% of the frame area. (ii) *Action-target augmentation.* We additively perturb every target pose in the action chunk with Gaussian noise— $\sigma_{\text{pos}}=1$ mm on translation and $\sigma_{\text{rot}}=0.5^\circ$ on rotation—before

the flow-matching loss is computed, which regularizes the velocity field against small tracking noise in the labels. (iii) *Temporal augmentation*. With $p=0.5$ we apply *sub-step interpolation*: adjacent state/action frames are linearly blended at a random $\alpha \in [0, 1]$, effectively densifying the temporal grid at no extra collection cost.

G Inference Details

G.1 Deployment Pipeline

Nothing inside the policy changes when it moves from human video to a robot: it consumes the same two inputs as in training—one embodiment-agnostic RGB image and a set of ICT tokens—and only the modules that *produce* those inputs are swapped.

Reference frame. At deployment the camera is static, so its frame serves directly as the episode’s reference frame REF: every entity pose and every predicted action is expressed in it, whichever camera is mounted. On human video the camera moves with the head, so we instead freeze the camera pose at the first manipulation frame and use it as REF. Each demonstration thus contributes its own reference frame, and a task’s demonstrations span the head poses a person naturally occupies at a table; a new camera placement is another draw from that distribution rather than an out-of-distribution input.

Observation assembly. Detection, segmentation, and orientation estimation run *once* per episode, from one text prompt per object. The robot-side frontend is the training frontend with triangulation replaced by depth back-projection, and needs neither SLAM nor the short scene sweep used to initialize human episodes. Every control cycle we capture an RGB-D frame, segment and inpaint the robot arm and render the virtual gripper in its place, and read arm forward kinematics and the gripper state to obtain T_{ee} and g . We then assemble each token as in Eq. (1): ${}^{\text{REF}}T_E$ in the static camera frame, ${}^E T_{LH}$ and ${}^E T_{RH}$ from the gripper pose, and g from the gripper opening. Latching carries over unchanged from training (Sec. 3.3), with the end effector in place of the hand: from grasp onset the object pose is propagated by the gripper rather than re-estimated, assuming no in-hand manipulation: a grasped object keeps a fixed pose relative to the gripper. This keeps the object token correct under occlusion, but a physical slip out of the gripper still ends the episode in failure.

Control loop. The policy integrates the learned flow field into a 50-step action chunk, decodes camera-frame end-effector targets, and maps them to arm commands through the hand-eye extrinsic. The loop is synchronous at ~ 10 Hz on a single RTX 4090, so each plan acts on a frame at most ~ 100 ms old while the predicted action chunk executes continuously between plans. Token assembly is negligible; runtime is dominated by arm inpainting and segmentation.

Requirements. The only per-rig setup is camera intrinsics and one hand-eye extrinsic per arm, both obtained once and reused across tasks. We use no CAD models, no manual initialization, no online object tracking, and no depth inside the policy—depth serves the frontend only. Hardware-agnostic therefore means no robot demonstrations and no task-specific calibration: the same checkpoint drives a different arm or camera once its extrinsic is known

G.2 Robot Inference Setup

Apart from the zero-shot generalization study (Sec. 4.3), all real-world experiments in the main paper use the single inference setup shown in Fig. 13: two Trossen WidowX AI arms mounted side-by-side on a shared workbench, forming a bi-manual platform that handles both single-arm and two-arm tasks without any hardware change between tasks. Each WidowX AI arm is a 6-DoF parallel-jaw manipulator with a ~ 1.5 kg payload at full reach and ± 1 mm end-effector repeatability. For visual input we use a single Intel RealSense D405 mounted top-down above the workspace; its RGB stream is the sole observation consumed by HumanEgo. Each WidowX AI arm also ships with a built-in wrist camera, but we deliberately do not use it for HumanEgo: the robot-teleoperation ACT baseline [28] in Sec. 4.1, in contrast, does consume the wrist cameras as part of its standard observation interface.



Fig. 13: Robot inference setup.

G.3 Flow Matching Rollout and Control

ODE rollout. At test time we integrate the learned velocity field with a fixed-step Euler solver using 20 inference steps: starting from a noise sample $\mathbf{x}_0 \sim \mathcal{N}(0, I)$ drawn once at policy load time, we iterate $\mathbf{x}_{t+\Delta t} \leftarrow \mathbf{x}_t + v_\theta(\mathbf{x}_t, t, s_t) \Delta t$ with $\Delta t = 1/20$, yielding a $K=50$ -step bimanual action chunk in one forward pass per re-plan. Predictions are unpacked dimension-wise into per-hand position, 6D rotation, and grasp logit, with positions denormalized by the dataset mean/std, rotations projected back to $\text{SO}(3)$ via *normalize-then-Gram-Schmidt* on the 6D representation, and grasps passed through a sigmoid.

Action chunking and control. The controller re-plans at every cycle (10 Hz), keeping at most one prediction in history and executing one action per cycle. A step stride of 2 sub-samples the chunk so the effective executed rate is 5 Hz, and a look-ahead offset of 25 steps lets the controller query the chunk slightly ahead of the current execution index to mask planning latency. For grasp we use an any-over-horizon rule: the gripper closes as soon as *any* step in the current chunk predicts a grasp probability above 0.6; an optional grasp-latch mode additionally locks the gripper closed after the first grasp event to prevent accidental mid-task releases.

Smoothing and safety. To hide small noise in the predicted $\text{SE}(3)$ stream we apply an EMA on positions ($\alpha=0.5$) and quaternion SLERP on rotations before streaming targets to the arms, and a trajectory-overlap blend (smoothing parameter 12) to avoid jerky starts/stops between consecutive chunks. Finally, a safety cage limits each per-cycle target displacement to ≤ 0.08 m in position and ≤ 0.02 rad in rotation to guard against sudden outliers; we did not observe any safety-cage clamp during normal rollouts in our experiments.

H Additional Experiments Analysis

H.1 Hand Tracking Method Study

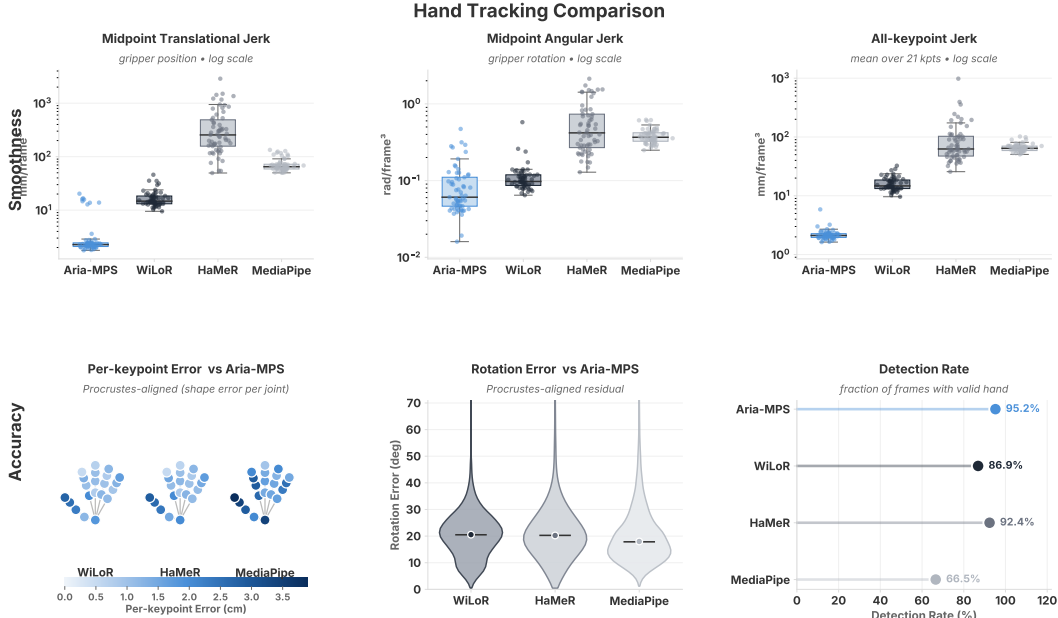


Fig. 14: **Hand tracking comparison on Serve Bread** (45 demonstrations, ~ 45 k frames). *Top*—*Smoothness*: per-frame jerk of the gripper midpoint (translational and angular) and of all 21 keypoints (lower is better, log scale). *Bottom*—*Accuracy vs. Aria-MPS*: per-keypoint shape error after Procrustes alignment, residual rotation error after subtracting the systematic frame offset, and fraction of frames with a valid hand detection.

Setup. ICT consumes 3D hand keypoints as input, so the quality of the upstream hand tracker directly affects what the policy can learn. We isolate this dependency on Serve Bread by holding everything else constant—the same 45 demonstrations (30 min total), the same HumanEgo

architecture, the same training recipe—and varying only the hand-tracking module that produces the action labels. We compare four trackers spanning the dominant design choices in the literature: (1) **Aria-MPS** [7], our default, which fuses the two wide-FoV *monochrome SLAM* cameras with the on-device IMU through Meta’s MPS pipeline to recover metric 3D keypoints—note that the central RGB camera is used only for video logging, not for hand tracking; (2) **WiLoR** [65], a transformer that regresses MANO parameters from a single RGB crop per frame; (3) **HaMeR** [66], a strong monocular RGB estimator that also predicts MANO parameters but processes frames independently—temporal and world-frame extensions of this family have since been proposed [67, 68]; and (4) **MediaPipe** [69], a lightweight monocular RGB pipeline whose 3D outputs are root-relative :

Beyond these vision-only trackers, instrumented alternatives recover hand pose from multi-modal sensing gloves [70] or 6-axis IMU systems [71], but require specialized hardware on the demonstrator and fall outside our zero-instrumentation collection setup. For each tracker we re-run the data preprocessing, train HumanEgo from scratch, and evaluate 40 real-world trials on Serve Bread (Fig. 15). Aria-MPS is treated as the reference for all alignment-style metrics in Fig. 14.

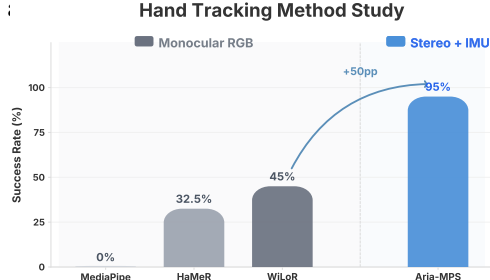


Fig. 15: **Hand Tracking Method Study.**

Results. Fig. 15 shows real-world success collapsing by $95 \rightarrow 45 \rightarrow 32.5 \rightarrow 0\%$ as we move from stereo Aria-MPS to monocular trackers, and Fig. 14 explains why. We highlight four observations.

Stereo depth is decisive for downstream success. Real-world success drops from 95 % (Aria-MPS) to at most 45 % (WiLoR) the moment we replace stereo with monocular RGB. Monocular networks are inherently scale-ambiguous along the depth axis, producing a 5–11 cm systematic depth offset that propagates directly into the ICT reference frame, so the policy never learns a consistent grasp.

Smoothness and tracking persistence—not pose accuracy per se—separate the surviving base-lines. After Procrustes alignment the per-keypoint residual error is nearly identical for HaMeR and WiLoR (1.4 cm vs. 1.4 cm; both dominated by the shared MANO inductive bias) and only modestly worse for MediaPipe (2.2 cm). Yet WiLoR (45 %) clearly beats HaMeR (32.5 %). The gap is explained by the smoothness panels: WiLoR’s gripper-midpoint jerk is more than an order of magnitude lower than HaMeR’s—its per-frame predictions happen to be far more temporally stable—and its detection rate is 86.9 % vs. MediaPipe’s 66.5 %. A jittery or intermittently missing trajectory teaches the policy incoherent action labels.

The MANO prior is a double-edged sword. HaMeR and WiLoR show essentially identical per-keypoint Procrustes residuals (Fig. 14, bottom-left): the colored patterns on the two hand skeletons are visually indistinguishable. This is not a coincidence—both networks regress MANO pose parameters and inherit the same canonical bone proportions. The shared shape prior means their residual shape errors are correlated by construction, while methods without a parametric constraint (Aria-MPS, MediaPipe) deviate in different ways. This explains why pose accuracy alone is a poor predictor of downstream policy success.

MediaPipe fails entirely. With 0 % real-world success and only 66.5 % detection rate, MediaPipe cannot produce coherent action labels even with the ICT representation absorbing some noise. Its 3D output is root-relative and must be lifted with the camera depth, and the lift collapses the hand thickness to ~ 7 cm (vs. ~ 16 cm for Aria), flattening the pose information that ICT relies on.

Together these results point to a clear practical message: invest in the perception frontend. A more accurate hand tracker—especially one that exploits stereo or learned depth—is the highest-leverage upgrade for any policy that operates on hand-derived spatial tokens.

H.2 Human-Robot Co-Training Study

Setup. The data-efficiency study (Sec. 4.2) showed that human demonstrations are roughly $3.75\times$ more sample-efficient than robot teleoperation *at the same total collection time*. Here we ask the

complementary question: when both modalities are available, what is the best mixing ratio? We hold the total collection time fixed at 30 min and vary the fraction of *human* data from 0 % (pure robot teleoperation, ~ 45 teleop episodes) to 100 % (pure human egocentric video, ~ 45 egocentric demos) in 25-pp steps. Each batch is sampled with the corresponding ratio, so the policy sees both modalities in the intended proportion at every gradient step. The architecture, optimizer, schedule, and number of training steps are kept identical across the five conditions; only the data mixture changes. We evaluate 40 real-world trials on Serve Bread per condition.

Results. Real-world success increases *monotonically* as the human-data ratio grows: $65 \rightarrow 72.5 \rightarrow 77.5 \rightarrow 90 \rightarrow 95$ % for human ratios of 0/25/50/75/100 % (Fig. 16). The pure-human policy improves over the pure-robot policy by +30 pp—a gap larger than that between most of our baselines. We extract two main findings.

Even a small slice of human data dominates.

Replacing just 25 % of the robot teleop with egocentric video already lifts success from 65 % to 72.5 % (+7.5 pp), even though the absolute amount of robot data drops from 30 min to 22.5 min in that condition. A naive “more data is always better” view would predict the opposite: less robot data should hurt. Instead the policy improves, indicating that the marginal robot minutes contribute less signal than the marginal human minutes. This reproduces the data-efficiency conclusion (Sec. 4.2) at the level of the gradient step: the policy is happier learning from a small amount of clean egocentric video than from a large amount of teleop trajectory.

The more human data, the better—no co-training sweet spot. Across all four transitions the curve only goes up, and the pure human policy is the global maximum. We find no “sweet spot” where mixing in robot data outperforms the human-only condition—in fact 75/25 (90 %) is already 5 pp below 100/0 (95 %), so adding even 7.5 min of robot teleoperation actively erodes a 22.5 min human dataset. We attribute this to the higher per-minute information density of human demonstrations documented in Sec. 4.2 (Fig. 6): human videos exhibit higher signal-to-noise ratio, an order-of-magnitude smoother trajectories, near-zero idle time, and broader spatial coverage than robot teleoperation. At a fixed compute and time budget, the policy is best served by spending every minute on human data. Combined with the embodiment-invariance of ICT, this means a practitioner deploying HumanEgo should not invest in robot teleoperation at all—the same budget collected as egocentric human video yields a strictly better policy.

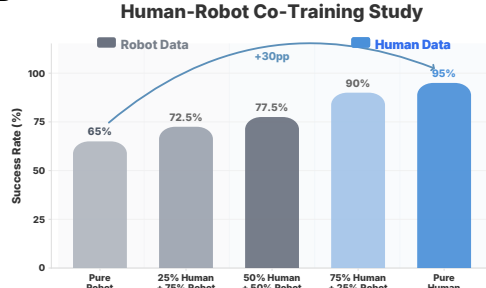


Fig. 16: **Human-Robot Co-Training Study.**

I Hyperparameters

Table 1 consolidates every hyperparameter value that appears in the main paper and the appendix, grouped by pipeline stage. Unless explicitly noted, a single value is used across all four tasks, all training runs, and all real-world trials.

Table 1: **All hyperparameters used in HumanEgo.** Values are shared across the four tasks unless noted.

Parameter	Value
<i>Data Collection (App. D.1, App. D.2)</i>	
Demonstrations per task	45
Total human-video time per task	30 min
RGB stream rate / resolution	30 fps / 2 MP
SLAM cameras: count, rate, resolution	2, 30 fps, VGA
Eye-tracking cameras: count, rate, resolution	2, 10 fps, QVGA
IMU rates	1000 Hz, 800 Hz
Pre-episode scene-sweep duration	~ 1 s
Scene-sweep frame count	~ 30

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Table 1 – continued from previous page

Parameter	Value
<i>Phase Detection (App. E.2)</i>	
Head linear-speed stop threshold v_{stop}	0.03 m/s
Head angular-speed stop threshold w_{stop}	0.15 rad/s
Minimum stop-hold duration	15 frames
Rotate trigger w_{rot}	0.10 rad/s
Rotate max linear speed $v_{\text{rot,max}}$	0.08 m/s
Transition buffer	10 frames
Hand-velocity demotion threshold v_{hand}	0.15 m/s
Hand-velocity averaging window	5 frames
Finished-stop length	30 frames
<i>Hand-to-Gripper Retargeting (App. E.3)</i>	
Retarget keypoints per hand	5
MPS confidence threshold	0.8
Minimum detection segment	30 frames
Max gap for interpolation	10 frames
Savitzky–Golay window	21 frames
Savitzky–Golay polynomial order	2
EMA smoothing factor $\alpha_x = \alpha_y$	0.15
<i>Policy Network (App. F.1)</i>	
Transformer layers	6
Transformer attention heads	8
Transformer embedding dim	384
Dropout	0.05
RGB patch size	16×16
RGB input resolution	240×320
ICT token dim	29
Prediction horizon K	50
<i>Losses (App. F.1)</i>	
Position weight w_p	5
Rotation weight w_r	1
Grasp weight w_g	10
Object-motion weight λ_{OM} (pos / rot)	$0.5 w_p / 0.5 w_r$
2D-trace weight λ_{2D}	20
Latent-consistency weight λ_{LC}	[0.1, 1.0]
<i>Optimization (App. F.1)</i>	
Optimizer	AdamW
Base learning rate	1×10^{-4}
Warmup steps	200
Min-LR ratio	0.05
Batch size	32
Epochs	400
Gradient-norm clip	1.0
EMA decay	0.999
<i>Data Augmentation (App. F.1)</i>	
Photometric jitter probability	0.8
Brightness / contrast delta	$\pm 0.20 / \pm 0.20$
Gamma delta	± 0.15
Pixel noise σ	0.02
Grayscale probability	0.1
HSV hue jitter	± 10
HSV saturation range	[0.6, 1.4]
Random resized crop probability	0.5
Scale range	[0.7, 1.0]
Aspect-ratio range	[0.9, 1.1]
Gaussian blur probability	0.15
Kernel size	3×3
Random erasing probability	0.5
Number of holes	3–8

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Table 1 – continued from previous page

Parameter	Value
Per-hole area	5–20%
Target position noise σ_{pos}	1 mm
Target rotation noise σ_{rot}	0.5°
Sub-step interpolation probability	0.5
<i>Hardware (App. G.2)</i>	
Robot arms	2× Trossen WidowX AI
WidowX AI DoF	6
WidowX AI payload at full reach	~1.5 kg
WidowX AI end-effector repeatability	±1 mm
Inference RGB camera	Intel RealSense D405 (top-mounted)
<i>Inference (App. G.3)</i>	
Euler ODE steps	20
Step size Δt	1/20
Executed action-chunk length	$K = 50$
Control-loop frequency	10 Hz
Action-step stride	2
Effective executed rate	5 Hz
Look-ahead offset	25 steps
Grasp probability threshold	0.6
Position smoothing EMA α	0.5
Rotation smoothing	quaternion SLERP
Trajectory-overlap smoothing parameter	12
Safety cage: max position step	0.08 m
Safety cage: max rotation step	0.02 rad
<i>Robot Teleoperation for ACT Baseline</i>	
Teleop control frequency (leader → follower)	200 Hz
Teleop recording frequency	30 Hz
Leader → follower EMA α	0.5
Top camera resolution	640 × 480
Wrist camera resolution	320 × 240
Action space	7-DoF joint positions
Proprioception dim	7
<i>ACT Baseline Training</i>	
Visual backbone	ResNet-18 (pretrained)
Embedding dim	256
Attention heads	8
Encoder / decoder layers	4 / 1
Feed-forward dim	2048
CVAE latent dim	32
Dropout	0.1
Prediction horizon K	50
Image input resolution	240 × 320
Batch size	24
Epochs	400
Base learning rate	1×10^{-4}
Weight decay	1×10^{-2}
Warmup steps	500
EMA decay	0.999
Action L1 weight w_{pos}	5.0
CVAE KL loss weight (annealed to)	10.0